

CONTACT

+39 329 087 5155

nicolaghibulet@gmail.com

linkedin.com/in/nicola-ghibulet-57ab7831a

Parma, Italy

Born 19 December 2002

SKILLS

Programming languages
Advanced: C/C++, Java, Python
Intermediate: SQL, PHP, JavaScript
Basic: C#, Assembly, Prolog

Machine Learning
PyTorch, Graph Neural Networks, NumPy, Pandas

Systems and tools
Linux and Bash, Git and GitHub, Windows, university HPC cluster (VPN, command line)

Foundations
Algorithms and data structures, parallel programming, databases, networking, operating systems, software engineering

LANGUAGES

Italian – native

Romanian – fluent

English – B2 (certified)

Russian – basic comprehension

INTERESTS

Competitive swimming (8 years), freediving, gym, multi-day mountain hiking and bivouacking

Nicolae Ghibulet

MSc Student in Computer Science

PROFILE

Computer Science graduate from the University of Parma (101/110), now enrolled in the MSc in Computer Science. During my thesis internship I modified the algorithmic core of GIST, a Graph Neural Network framework for spatial transcriptomics, and validated the new version across 17 experiments. Solid skills in C/C++, Java and Python, with hands-on parallel programming on an HPC cluster. Looking for a research internship or a Master's thesis project abroad.

EDUCATION

MSc in Computer Science 2026 – present
University of Parma, Italy

BSc in Computer Science 2022 – 2026
University of Parma, Italy · Final grade: **101/110**

Thesis: "Application of Quantized Divergence in GNNs for Clustering" – supervisor Prof. Vincenzo Bonnici

Key courses (out of 30): Programming Methodologies (30 cum laude), Programming Fundamentals (29), Algorithms and Data Structures (28), Parallel Programming and HPC (26), Mathematical Analysis (30)

Technical High School Diploma – Computer Science 2016 – 2022
ITIS, Parma, Italy · Grade: **99/100**

EXPERIENCE

Thesis Internship – Graph Machine Learning 2025 – 2026
Dept. of Mathematical, Physical and Computer Sciences, University of Parma

- Reviewed the state of the art in GNNs (GCN, GAT, GraphSAGE, GIN, VGAE) and graph contrastive learning.
- Modified the structural loss of GIST (University of Verona): KL divergence computed on a quantized adjacency matrix, working with the model's first author.
- Validated on 6 10x Visium datasets (17 experiments): average Silhouette Spatial Score improved by +0.031, with negligible computational overhead.
- Python, PyTorch · github.com/synbionics/GIST_nkl

Competitive Swimming Coach 2020 – present
Ego Village, Collecchio (Parma)

Training young athletes, planning sessions and leading the team.

Lifeguard (seasonal) 2024 – present
Ego Village, Collecchio (Parma)

Short Internship June 2021
Sopra Steria, Collecchio (Parma)

Studied and applied Python libraries (NumPy, Pandas, web automation) in a personal project.